Departamento Inteligencia Artificial

Artificial Intelligence Master

Universidad Politécnica de Madrid

Search Engines & Machine Learning Ranking

Prof. David Pérez del Rey

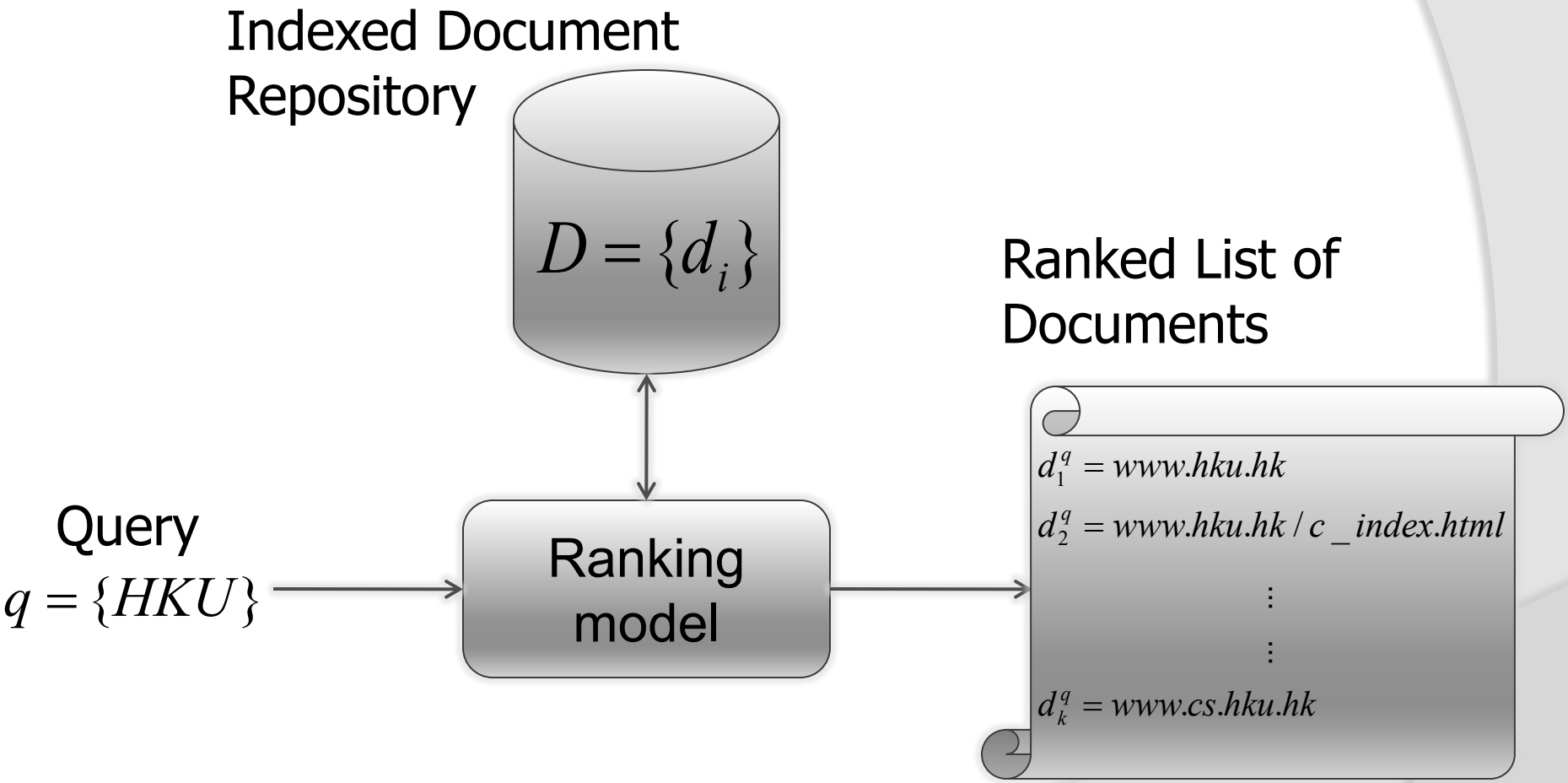
dperezdelrey@fi.upm.es

Room 2106

Overview

- Search Engines
- Machine Learning Ranking
  - Pointwise approach
  - Pairwise approach
  - Listwise approach

Inside Search Engine



Liu, Tie-Yan. "Learning to rank for information retrieval." *Foundations and Trends® in Information Retrieval* 3.3 (2009): 225-331.

Conventional Ranking Models

Category	Example Models
Similarity-based models	Boolean Model
	Vector Space Model
	Latent Semantic Indexing
	… …
Probabilistic models	BM25 Model
	Language Model for IR
	… …
Hyperlink-based models	HITS
	PageRank
	… …

Machine Learning Can Help

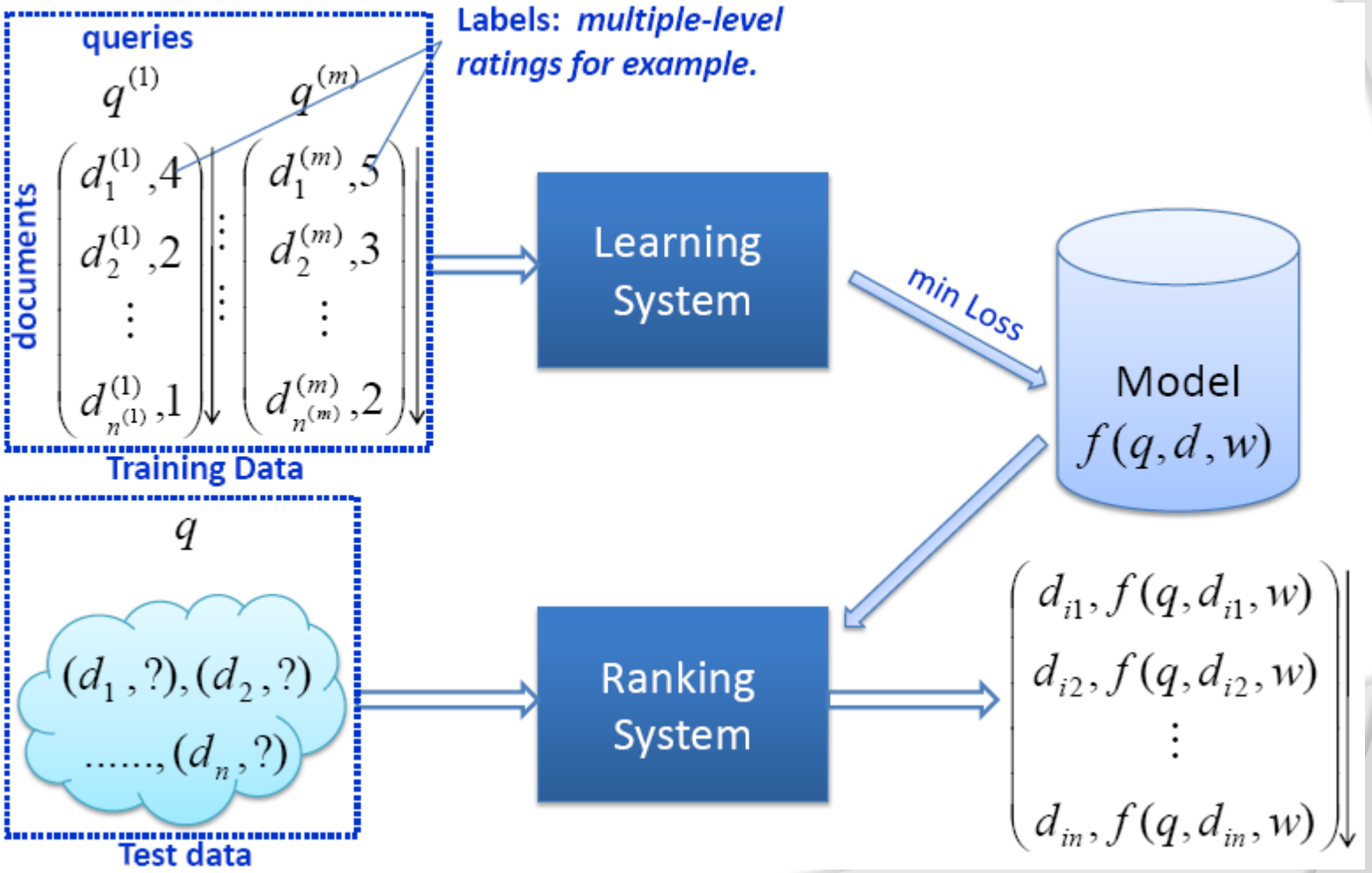
- Machine learning can
  - Automatically tune parameters
  - Combine multiple evidences
  - Avoid over-fitting
  - Customize retrieval
- "Learning to Rank"
  - Use machine learning techniques to train the ranking model

Machine Learning Ranking

- Information Retrieval + Machine Learning → Machine-learned ranking (MLR)

- Advantages:
  - The ranking model adapts to the context (user behavior, information nature, …)
  - Easy addition of new documents to the set
  - Benefit from the advances of Machine Learning field

Learning to Rank Foundations Framework



Departamento Inteligencia Artificial

Widely-used Judgment

- Binary judgment
  - Relevant vs. Irrelevant
- Multi-level ratings
  - Perfect > Excellent > Good > Fair > Bad
- Pairwise preferences
  - Document *A* is more relevant than document *B* w.r.t. query *q*

Learning to Rank Foundations
Categorization: Basic Unit of Learning

- Pointwise
  - Input: single documents
  - Output: scores or class labels
- Pairwise
  - Input: document pairs
  - Output: partial order preference
- Listwise
  - Input: document collections
  - Output: ranked document list

Learning to Rank Foundations
Training data

- Binary classified query results
  - Documents labeled as relevant/not relevant to the query

query

Result 1	Relevant
Result 2	Irrelevant
Result 3	Irrelevant
Result 4	Relevant

- Click-through data
  - Triplets: query, ranking presented to the user, set of documents clicked

query

Result 1
Result 2
Result 3 (clicked)
Result 4 (clicked)

- Ranked query results
  - Ranked results to a given query

query

Result 1	2nd
Result 2	4th
Result 3	1st
Result 4	3rd

Learning to Rank Foundations
Ordinal Regression

- Input space
  - Features of a single document (w.r.t. a query): $X \in R^T$
- Output space
  - **Ordered** categories: $Y \in \{c_1 \prec c_2 \prec \ldots \prec c_K\}$

$$q_i \quad \begin{pmatrix} x_1^{(i)} \\ x_2^{(i)} \\ \vdots \\ x_{M^{(i)}}^{(i)} \end{pmatrix} \Rightarrow \begin{array}{c} q_i \\ \{(x_1^{(i)}, c_4), (x_2^{(i)}, c_3), \ldots, (x_{M^{(i)}}^{(i)}, c_1)\} \\ c_1 \prec c_2 \prec c_3 \prec c_4 \end{array}$$

Pointwise approach: Staged Logistic Regression

- Training data: binary
- Approach:
  - $f(query, document) \rightarrow relevance\ score$
  - Order results by the $relevance\ score$
- E.g. Staged Logistic Regression ranker (Cooper, 1992)
  - Goal: Estimate the probability of a document to be useful to the user for a given query using a logistic regression model

Cooper, William S., Fredric C. Gey, and Daniel P. Dabney. "Probabilistic retrieval based on staged logistic regression." Proceedings of the 15th annual international ACM SIGIR conference on Research and development in information retrieval. 1992.

Pairwise approach: Ranking SVM

- Training data: Click-through
- Approach:
  - $f(query, document_1, document_2) \rightarrow relevance\ comparison$
  - Order results using f
- E.g. Ranking SVM (Joachims, 2002)
  - Goal: Use click-through data to build an SVM model that predicts, from a pair of documents, the most relevant one

Joachims, T. (2002, July). Optimizing search engines using clickthrough data. In *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 133-142). ACM

Listwise approach: AdaRank

- Training data: Ranked results to given queries
- Approach:
  - $f(query, \boldsymbol{S}) \rightarrow \boldsymbol{R}$
  - $\boldsymbol{S} = set\ of\ documents$
  - $\boldsymbol{R} = list\ of\ documents\ ordered\ by\ relevance$
- E.g. AdaRank (Xu & Li, 2007)
  - Goal: Train an AdaBoost model to rank the results of given queries

Xu, J., & Li, H. (2007, July). Adarank: a boosting algorithm for information retrieval. In *Proceedings of the 30th annual international ACM SIGIR conference on Research and development in information retrieval* (pp. 391-398). ACM.

Assignment (Groups 2-4 students)

1. Choose a type of MLR approach
2. Understand how it Works
3. (completely independent task to understand LOINC) Submit the LOINC mapping assignment during the lecture on 25th February
4. Given a list of documents for 3 queries ("glucose in blood", "bilirubin in plasma", "White blood cells count"), build the appropiate training set for the chosen approach

Optional to get over 60% of the grade

5. Implement model (you can use public libraries)
6. Extend dataset in # terms
7. Extend dataset in queries
8. Other creative analysis / improvements / developments

The Objective is to Understand and be able to Explain the process, not to get the best results

16th March 2026 – Moodle, submit a .zip file with:

- Training dataset files
- Report stating including contribution from each member (in .pdf) **up to 4 pages**
- Presentation
- Any additional material used

18th March 2026 – Presence **mandatory** – **Presentation** of the MLR approach (up to 10mins)

Departamento Inteligencia Artificial

Artificial Intelligence Master

Universidad Politécnica de Madrid

Questions?

Prof. David Pérez del Rey

dperezdelrey@fi.upm.es

Room 2104 / 5004

Tel: +34 91 336 74 45 / 74 67